

Deep Networks for Monitoring Waterway Traffic in the Mekong Delta

Thanh-Nghi Do^{1,2}, Minh-Thu Tran-Nguyen¹
Thanh-Tri Trang¹, and Tri-Thuc Vo¹

¹ College of Information Technology
Can Tho University, 92000-Cantho, Vietnam

² UMI UMMISCO 209 (IRD/UPMC)
Sorbonne University, Pierre and Marie Curie University - Paris 6, France
`dtngchi@cit.ctu.edu.vn`

Abstract. Our investigation aims at training deep networks for monitoring waterway traffic means on the rivers in the Mekong Delta. We collected the real videos of the waterway traffic, and then tagging the five most popular means in frames extracted from the videos, making an image dataset. We propose to train recent deep network models such as YOLO v4 (You only look once), RetinaNet and EfficientDet on this image dataset to detect the five most popular means in the videos. The numerical test results show that YOLO v4 gives highest accuracy than two other methods, including RetinaNet and EfficientDet. YOLO v4 achieves the performances on the testset with a precision of 91%, a recall of 98%, F1-score of 94% and mean average precision (mAP@0.50) of 97.51%.

Keywords: Waterway traffic means · Deep network · YOLO v4 · RetinaNet · EfficientDet.

1 Introduction

The Mekong Delta in southern Vietnam is a vast system of rivers, swamps, home to floating markets, and villages surrounded by rice paddies. Waterway traffic means are the main transportation of the region. Therefore, monitoring waterway means is the promising research to avoid traffic jams.

Recently, computer vision is a field that gets more attention due to its applications. One of the most common problems related to computer vision is object detection that related to the ability of computer systems can locate objects in an image and identify them. Along with the deep neural network (called AlexNet proposed by [13]), this has promoted the development of deep learning networks to extract information from images and videos based on Convolutional Neural Networks (CNNs [14]). In addition, CNNs have proven to be one of the best approaches for image classification and image recognition [10]. That is why we propose to train deep neural networks for monitoring waterway traffic means on the rivers in the Mekong Delta.

In order to build the system for monitoring waterway traffic means, we firstly collect the real videos of the waterway traffic, and then the frames extracted from the videos are manually drawn by bounding boxes around the five most popular means, including Canoe, Composite Canoe, Boat, Ship and Barge. We obtain an image dataset. After that, we propose to train three recent deep network models, such as You only look once (YOLO v4 [2]), RetinaNet [16] and EfficientDet [28] to detect the five most popular means in the videos. The empirical test results on our real image dataset show that YOLO v4 achieves higher accuracy and faster training time than two other methods, including RetinaNet and EfficientDet. YOLO v4 gives the performances on the testset with a precision of 91%, a recall of 98%, F1-score of 94% and mean average precision (mAP@0.50) of 97.51%.

The remainder of this paper is structured as follows. Section 2 briefly presents our proposal for monitoring waterway traffic means with deep networks, including YOLO v4, RetinaNet and EfficientDet. Section 3 shows the experimental results. The conclusion is presented in section 4.

2 Training deep networks for monitoring waterway traffic means in the Mekong Delta

Our system for monitoring waterway traffic means follows the usual framework of the image classification. Building this system involves three main works as follows:

1. collecting the dataset of images,
2. extracting visual features from images and representing them,
3. training classifiers.

2.1 Data collection of waterway traffic means

Therefore, we start with the collection of the waterway traffic images. We used the camera of the Samsung Galaxy A10 (with 13 MP, f/1.9, 28mm wide, AF, 1080p@30fps) to record the waterway traffic on rivers in the Mekong Delta region, closed to floating markets. From these videos, we extracted 4480 images. And then the images are manually drawn by bounding boxes around the five most popular means, including Canoe, Composite Canoe, Boat, Ship and Barge. Fig. 1 presents an sample about five waterway traffic means.



Fig. 1. Categories of waterway traffic means

We divided the image dataset into two sub-sets: training set (to train the model) and test set (to evaluate the model) with the number of images 4027 and 453, respectively. Table 1 shows the description of the dataset with the number of five popular means in the training set and the test set.

Table 1. Description of dataset

No	Class	Trainset	Testset
1	Canoe	497	49
2	Composite Canoe	8,977	982
3	Boat	3,095	367
4	Ship	11,092	1,370
5	Barge	2,001	209
6	Total	25,662	2,977

2.2 Deep networks for monitoring waterway traffic means

The classical visual approaches perform the classification task of images via two key steps. The first one is to extract visual features from images and represent them, using handcrafted features including the scale-invariant feature transform (SIFT [19, 20]) and the bag-of-words model (BoW [3, 15, 26]), the histogram of oriented gradients (HOG [5]), the GIST [21]. Followed which, the second one is to train Support vector machines (SVM [29]) to classify images.

More recent approaches aim to train deep convolutional neural networks (CNNs [14]) to benefit from the ability to learn visual features (low-level, mid-level, high-level) from images and the softmax classifier in an unified framework. These approaches are widely used for the object detection task due to their performance. The object detection techniques are categorized into two genres: two-stage detectors and one-stage detectors. Two-stage detectors propose the potential bounding box candidates, and then a classifier only processes the region candidates. There are outstanding studies about object detection based on a two-stage approach such as Regions with CNN features (RCNN [8, 9]), Spatial Pyramid Pooling Networks (SPPNet [11]), Fast RCNN [7], Faster RCNN [25], and Feature Pyramid Networks [16]. One-stage detectors locate objects in images using a single deep neural network. The first one-stage detector in deep learning is considered You Only Look Once (YOLO) that was proposed by [22–24]. Liu et al. [18] proposed Single Shot MultiBox Detector (SSD) in 2016. Lin et al. [16] proposed the focal loss through RetinaNet to solve the foreground-background class imbalance. EfficientDet was proposed by Tan et al. [28], that is a one-stage detector by combine EfficientNet backbones, BiFPN and compound scaling.

We propose to train the three most recent deep network models, such as YOLO v4 [2], RetinaNet [16], and EfficientDet [28], for monitoring waterway traffic means, due to their performance in the object detection.

YOLO v4: Alexey Bochkovskiy proposed YOLO v4 to improve from an version of YOLO v3 [22]. The architecture of YOLO v4 consists of: CSPDarknet53 [30] playing a role as Backbone, Neck including SPP [11] and PAN [17], and YOLO v3 [22] (anchor based) as head. CSPDarknet53 is a CNN for object detection that is based on DarkNet-53 (53 convolutional Layers) and a CSPNet strategy (Cross Stage Partial Network). In YOLO v4, the SPP strategy (spatial pyramid pooling) is added the CSPDarknet53 because the receptive field is significantly increased. PANet (Path Aggregation Network) is employed in YOLO v4 as the parameter aggregation method from different backbone levels for different detector levels, compare with YOLO v3 used the FPN (Feature Pyramid Network [16]). The overall YOLO v4 architecture is presented in Fig. 2.

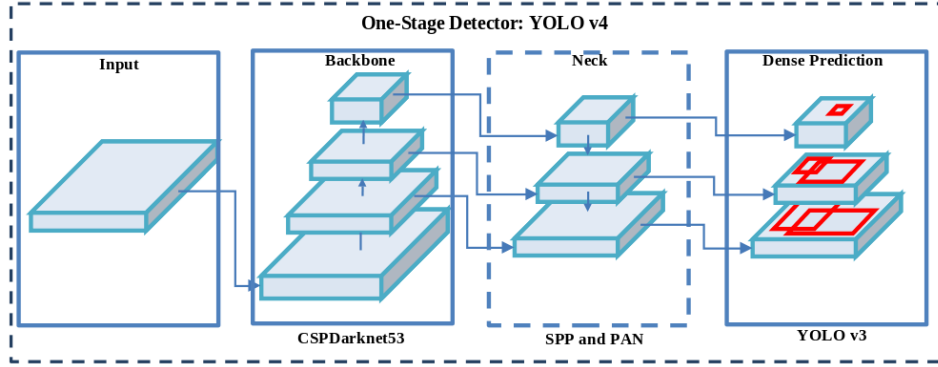


Fig. 2. YOLO v4 architecture

EfficientDet: Tan and his colleagues [28] designed the one stage detector paradigm called EfficientDet by linking EfficientNet backbones to a weighted bi-directional feature pyramid network (BiFPN) and combine with a customized compound scaling method [28]. Firstly, the authors observe that the efficiency of EfficientNets [27] is better achieved than previous commonly used backbones. Secondly, BiFPN enables easy and fast multiscale feature fusion. Thirdly, a compound scaling method is proposed for object detectors, that jointly scales up the resolution, depth, and width for all backbone, feature network, and box/class prediction networks. Based on the key optimization, EfficientDet consistently achieves both greater accuracy and better efficiency than previous object detection. Fig. 3 shows the overall architecture of EfficientDet.

RetinaNet: Tsung-Yi Lin and his colleagues proposed RetinaNet [16] to deal with the extreme foreground-background class imbalance that is the primary cause during training of one-stage object detectors. The main idea is to use the focal loss as illustrated in equation 1, where $\gamma > 0$ does reduce the relative loss for well-classified (q_i) and putting more focus on hard examples. Therefore, the training speed is compared to previous one-stage detectors while achieving

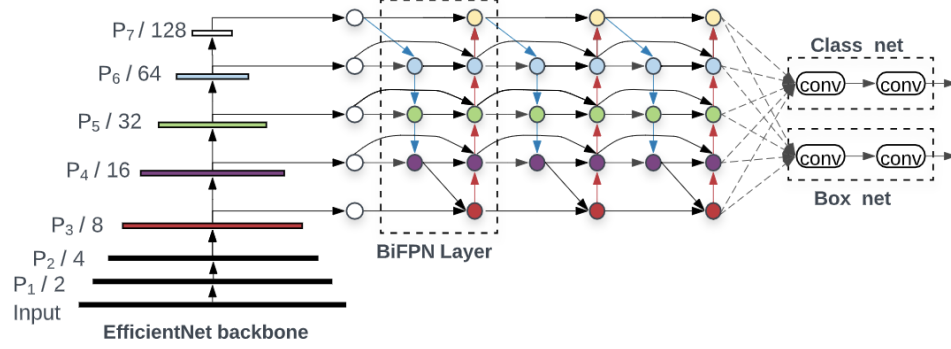


Fig. 3. EfficientDet architecture

higher accuracy and surpass with the accuracy of some existing state-of-the-art two-stage detectors like ResNet-101-C4 and Inception-ResNet-v2. The network architecture ResNet50-FPN (U-Net) (Fig. 4) is used to extract image features for the object detection.

$$FP(q) = -\alpha_i(1 - q_i)^\gamma \log(q_i) \quad (1)$$

where γ is focusing parameter and α is an α -balanced variant of the focal loss.

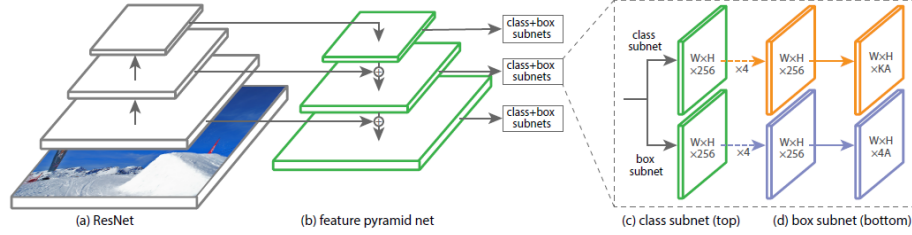


Fig. 4. RetinaNet architecture

3 Experimental results

In this section, we present experimental results of different deep networks for detecting waterway traffic means in the collected videos. For manual tagging the five waterway traffic means in images extracted from the videos, we implemented our own tagging tool in Python using library OpenCV [12]. We downloaded the

source of the YOLO v4 from [2] and compiling it with the custom configuration. We use the Keras implementation of RetinaNet object detection [6] and EfficientDet (Scalable and Efficient Object Detection [28]) implementation in Keras [4] and Tensorflow [1].

All experiments are conducted on a machine Linux Ubuntu 18.04, Intel(R) Core i7-4790 CPU, 3.6 GHz, 4 cores and 16 GB main memory and the Gigabyte GeForce RTX 2080Ti 11GB GDDR6, 4352 CUDA cores.

As described in section 2.1, we have the image dataset extracted from the waterway traffic videos on the rivers in the Mekong Delta. After manually drawn by bounding boxes around the five most popular means, including Canoe, Composite Canoe, Boat, Ship and Barge in images, we randomly split the image dataset into the trainset (4027 images) and testset (453 images). The description of the dataset is presented in Table 1. We use the trainset to train deep network models. Then, results are reported on the testset using the resulting deep network models.

3.1 Performance measurements

In the comparison of detection results obtained by the deep network models, we use the most popular performance measurements, including precision, recall, F1, Intersection over union (IoU), average precision (AP), and mean average precision (mAP).

- TP (True positive rate) measures the proportion of positives that are correctly identified.
- FN (False negative rate) is the proportion of positives that are mis-classified into negative by the model.
- FP (False positive rate) is the proportion of negatives that are mis-classified into positive by the model.

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (4)$$

The IoU metric measures the number of pixels common between the target (the ground-truth bounding box) and the prediction (the predicted bounding box) divided by the total number of pixels present across both.

$$IoU = \frac{area(B_p \cap B_{gt})}{area(B_p \cup B_{gt})} \quad (5)$$

where B_p is the predicted bounding box and B_{gt} is the ground-truth bounding box.

$$P_{interp} = \max_{r' \geq r} p(r') \quad (6)$$

$$AP = \sum_{i=1}^{n-1} (r_{i+1} - r_i) p_{interp}(r_i + 1)$$

where $(r_1, r_2, \dots, r_n)$ is recall levels $[0, 0.1, \dots, 1]$

$$mAP = \frac{mAP_{0.50} + mAP_{0.55} + \dots + mAP_{0.95}}{10} \quad (7)$$

where $(r_1, r_2, \dots, r_n)$ is recall levels $[0, 0.1, \dots, 1]$

3.2 Results

We obtain classification results obtained by deep network models in Table 2 and Figures 5, 6, 7, 8, 9 and 10. The highest accuracy is bold-faced and the second one is in italic.

In the comparison among deep network models, RetinaNet, YOLO v4 and EfficientDet, we can see that the RetinaNet and YOLO v4 models give highest average precision (AP) results (greater than 89%) with threshold IoU ≥ 0.5 . The YOLO v4 achieves the best results for 4/5 classes against RetinaNet.

The EfficientDet-0 ($\phi = 0$ is a hyper-parameter for scaling BiFPN width and depth in the EfficientDet network) and EfficientDet-1 ($\phi = 1$) models have lower AP than RetinaNet and YOLO v4. The EfficientDet models give AP of over 79% (except for Efficient-Det-0 for Canoe class being 50.91%).

Regarding the evaluation on mAP (COCO), the YOLO v4 model gives the highest accuracy of 97.51%, followed by the RetinaNet model, the EfficientDet-1 with accuracy of 96.28% and 93.39%, respectively.

Table 2. Classification results in terms of average precision for 5 waterway traffic means

Class	RetinaNet	YOLO v4	EfficientDet-0	EfficientDet-1
Canoe	<i>89.00</i>	94.10	50.91	85.79
Composite Canoe	<i>95.91</i>	95.94	79.06	88.29
Boat	<i>98.36</i>	99.23	94.48	94.17
Ship	99.48	99.21	97.11	<i>99.45</i>
Barge	98.67	<i>99.05</i>	98.70	99.23
mAP	<i>96.28</i>	97.51	84.05	93.39

More details on the results of the YOLO v4 model with threshold IoU 0.50, it achieves AP of over 94% for all waterway traffic means. In which 3 out of 5 classes have more than 99% AP accuracy corresponding Boat (99.23%), Barge (99.05%) and Ship (99.21%). Mean average precision mAP@0.50 reaches 97.51%.

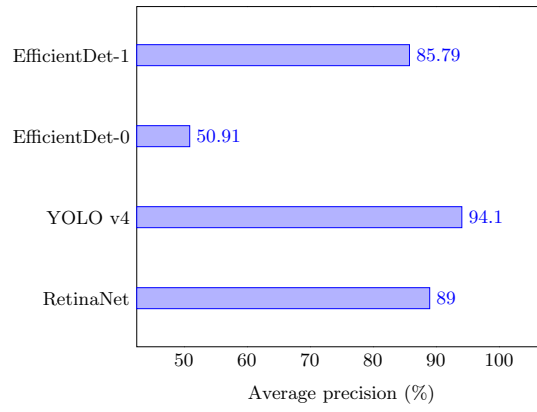


Fig. 5. Average precision for Canoe

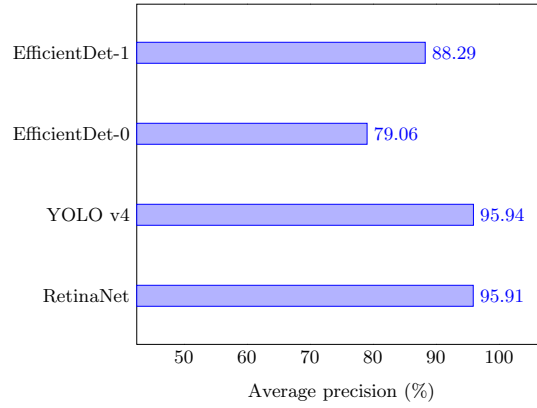


Fig. 6. Average precision for Composite Canoe

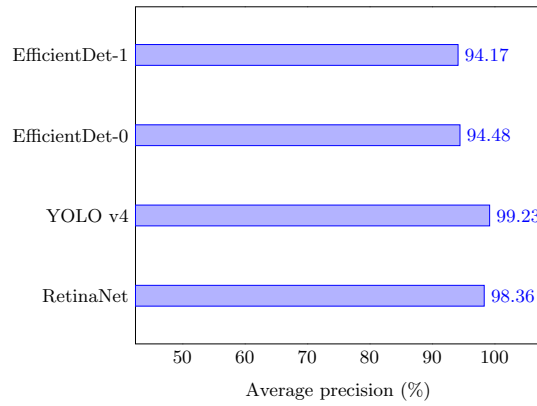


Fig. 7. Average precision for Boat

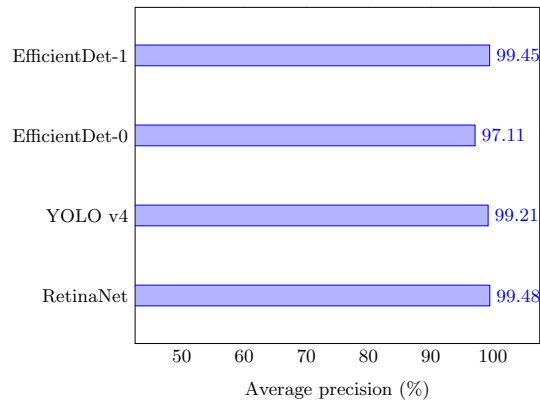
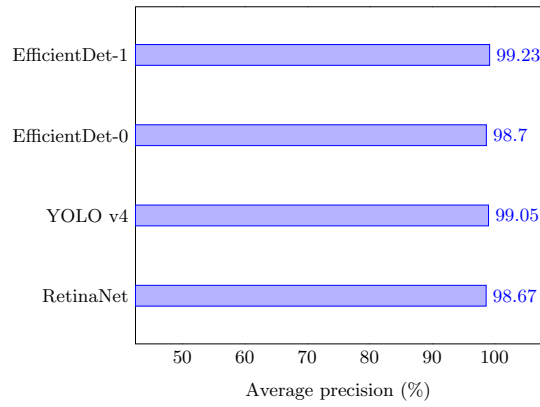
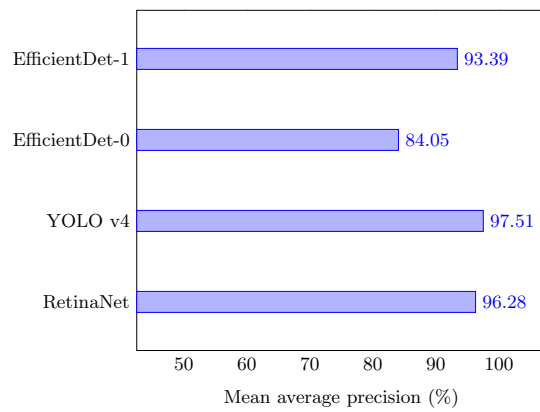
**Fig. 8.** Average precision for Ship**Fig. 9.** Average precision for Barge**Fig. 10.** Mean average precision



Fig. 11. A recognition result of waterway traffic means

The average IoU is 79.46%. The YOLO v4 gives a precision of 91%, a recall of 98%, F1-score of 94%. Figure 11 is a recognition result of YOLO v4.

4 Conclusion and future works

We have presented a proposal to monitor waterway traffic means on the rivers in the Mekong Delta through deep learning models. For this aim, we collected the real videos of the waterway traffic and manual tagging the five most popular means, including Canoe, Composite Canoe, Boat, Ship and Barge. After that, we trained three recent deep learning neural networks, such as YOLO v4, RetinaNet model, EfficientDet. The empirical test results show that the YOLO v4 model achieves the highest accuracy of 97.51% at the IoU threshold (0.5: 0.95) compared to accuracy of 96.28%, 93.39% obtained by RetinaNet model and EfficientDet, respectively. YOLO v4 gives the performances on the testset with a precision of 91%, a recall of 98%, F1-score of 94% and mean average precision (mAP@0.50) of 97.51%.

In the near future, we intend to provide more empirical test and compare with other algorithms. A promising future research aims to apply the YOLO v4 model to build a water traffic monitoring system on rivers as well as floating markets in the Mekong Delta.

Acknowledgments This work has received support from the College of Information Technology, Can Tho University. The authors would like to thank very much the Big Data and Mobile Computing Laboratory.

References

1. Abadi, M., Agarwal, A., Barham, P., Brevdo, E., Chen, Z., Citro, C., Corrado, G.S., Davis, A., Dean, J., Devin, M., Ghemawat, S., Goodfellow, I., Harp, A., Irving, G., Isard, M., Jia, Y., Jozefowicz, R., Kaiser, L., Kudlur, M., Levenberg, J., Mané, D., Monga, R., Moore, S., Murray, D., Olah, C., Schuster, M., Shlens, J., Steiner, B., Sutskever, I., Talwar, K., Tucker, P., Vanhoucke, V., Vasudevan, V., Viégas, F., Vinyals, O., Warden, P., Wattenberg, M., Wicke, M., Yu, Y., Zheng, X.: TensorFlow: Large-scale machine learning on heterogeneous systems (2015), <https://www.tensorflow.org/>, software available from tensorflow.org
2. Bochkovskiy, A., Wang, C.Y., Liao, H.: Yolov4: Optimal speed and accuracy of object detection. ArXiv **abs/2004.10934** (2020)
3. Bosch, A., Zisserman, A., Munoz, X.: Scene classification via pLSA. In: Proceedings of the European Conference on Computer Vision. pp. 517–530 (2006)
4. Chollet, F., et al.: Keras. <https://keras.io> (2015)
5. Dalal, N., Triggs, B.: Histograms of oriented gradients for human detection. In: Proceedings of the 2005 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR’05) - Volume 1 - Volume 01. pp. 886–893. IEEE Computer Society (2005)
6. Gaiser, H., de Vries, M., Lacatusu, V., vcarpani, Williamson, A., Liscio, E., Andrs, Henon, Y., jjiun, Gratie, C., Morariu, M., Ye), ., Zlocha, M., Weinstein, B., de Andrade, R.M., Conceio, P., Pacha, A., hannededvartsen, Shahrokhian, D., Fang, W., Clark, M., meagerYak, Jordal, I., Sande, M.V., Jin, Etienne-Meunier, Grigorev, A., Erhard, G., Ramos, E., Dowling, D.: fizyr/keras-retinanet 0.5.1 (Jun 2019). <https://doi.org/10.5281/zenodo.3250670>, <https://doi.org/10.5281/zenodo.3250670>
7. Girshick, R.: Fast r-cnn. In: 2015 IEEE International Conference on Computer Vision (ICCV). pp. 1440–1448 (2015). <https://doi.org/10.1109/ICCV.2015.169>
8. Girshick, R., Donahue, J., Darrell, T., Malik, J.: Rich feature hierarchies for accurate object detection and semantic segmentation. In: 2014 IEEE Conference on Computer Vision and Pattern Recognition. pp. 580–587 (2014). <https://doi.org/10.1109/CVPR.2014.81>
9. Girshick, R., Donahue, J., Darrell, T., Malik, J.: Region-based convolutional networks for accurate object detection and segmentation. IEEE Transactions on Pattern Analysis and Machine Intelligence **38**(1), 142–158 (2016). <https://doi.org/10.1109/TPAMI.2015.2437384>
10. Goodfellow, I., Bengio, Y., Courville, A.: Deep Learning. MIT Press (2016), <http://www.deeplearningbook.org>
11. He, K., Zhang, X., Ren, S., Sun, J.: Spatial pyramid pooling in deep convolutional networks for visual recognition. In: Fleet, D., Pajdla, T., Schiele, B., Tuytelaars, T. (eds.) Computer Vision – ECCV 2014. pp. 346–361. Springer International Publishing, Cham (2014)
12. Itseez: Open source computer vision library. <https://github.com/itseez/opencv> (2015)
13. Krizhevsky, A., Sutskever, I., Hinton, G.E.: Imagenet classification with deep convolutional neural networks. Commun. ACM **60**(6), 8490 (May 2017). <https://doi.org/10.1145/3065386>, <https://doi.org/10.1145/3065386>
14. LeCun, Y., Bottou, L., Bengio, Y., Haffner, P.: Gradient-based learning applied to document recognition. In: Proceedings of the IEEE. vol. 86, pp. 2278–2324 (1998)

15. Li, F., Perona, P.: A bayesian hierarchical model for learning natural scene categories. In: 2005 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR 2005), 20-26 June 2005, San Diego, CA, USA. pp. 524–531 (2005)
16. Lin, T.Y., Dollr, P., Girshick, R., He, K., Hariharan, B., Belongie, S.: Feature pyramid networks for object detection. In: 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 936–944 (2017). <https://doi.org/10.1109/CVPR.2017.106>
17. Liu, S., Qi, L., Qin, H., Shi, J., Jia, J.: Path aggregation network for instance segmentation. In: 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 8759–8768 (2018). <https://doi.org/10.1109/CVPR.2018.00913>
18. Liu, W., Anguelov, D., Erhan, D., Szegedy, C., Reed, S., Fu, C.Y., Berg, A.C.: Ssd: Single shot multibox detector. In: Leibe, B., Matas, J., Sebe, N., Welling, M. (eds.) *Computer Vision – ECCV 2016*. pp. 21–37. Springer International Publishing, Cham (2016)
19. Lowe, D.: Object recognition from local scale invariant features. In: *Proceedings of the 7th International Conference on Computer Vision*. pp. 1150–1157 (1999)
20. Lowe, D.: Distinctive image features from scale invariant keypoints. *International Journal of Computer Vision* pp. 91–110 (2004)
21. Oliva, A., Torralba, A.: Modeling the shape of the scene: A holistic representation of the spatial envelope. *International Journal of Computer Vision* **42**, 145–175 (2001)
22. Redmon, J., Divvala, S., Girshick, R., Farhadi, A.: You only look once: Unified, real-time object detection. In: 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 779–788 (2016). <https://doi.org/10.1109/CVPR.2016.91>
23. Redmon, J., Farhadi, A.: Yolo9000: Better, faster, stronger. In: 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 6517–6525 (2017). <https://doi.org/10.1109/CVPR.2017.690>
24. Redmon, J., Farhadi, A.: Yolov3: An incremental improvement. *ArXiv abs/1804.02767* (2018)
25. Ren, S., He, K., Girshick, R., Sun, J.: Faster r-cnn: Towards real-time object detection with region proposal networks. In: *Proceedings of the 28th International Conference on Neural Information Processing Systems - Volume 1*. p. 9199. NIPS’15, MIT Press, Cambridge, MA, USA (2015)
26. Sivic, J., Zisserman, A.: Video google: A text retrieval approach to object matching in videos. In: 9th IEEE International Conference on Computer Vision (ICCV 2003), 14-17 October 2003, Nice, France. pp. 1470–1477 (2003)
27. Tan, M., Le, Q.V.: Efficientnet: Rethinking model scaling for convolutional neural networks. *ArXiv abs/1905.11946* (2019)
28. Tan, M., Pang, R., Le, Q.V.: Efficientdet: Scalable and efficient object detection. In: 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 10778–10787 (2020). <https://doi.org/10.1109/CVPR42600.2020.01079>
29. Vapnik, V.: *The Nature of Statistical Learning Theory*. Springer-Verlag; 2nd edition (2000)
30. Wang, C.Y., Mark Liao, H.Y., Wu, Y.H., Chen, P.Y., Hsieh, J.W., Yeh, I.H.: Cspnet: A new backbone that can enhance learning capability of cnn. In: 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW). pp. 1571–1580 (2020). <https://doi.org/10.1109/CVPRW50498.2020.00203>